

StatsChitran: Mathematical and Practical Documentation

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1 Introduction

This vignette provides mathematical and practical documentation for selected functions in the **StatsChitran** package.

The aim is to document both the usage of the functions and the mathematical reasoning behind their implementation.

2 Linear modelling functions

2.1 The linreg function

This function `StatsChitran::linreg` involves fitting a linear regression model to a multivariate case while returning a message in case of multicollinearity

The model used is a non-weighted OLS

2.1.1 The univariate function

Start with a simple fit for a $y = ax + b + \epsilon$ case where $\epsilon \sim N(0, \sigma^2)$. Obviously the model in this case would be $f(x) = ax + b$

Then let $(x_i, y_i) \mid i \in [1, n]$ be the dataset. Then the the objective cost function involving the RSS according to OLS becomes

$$\mathcal{L}(a, b) = \sum_i \{y_i - ax_i - b\}^2 \quad (1)$$

$$(2)$$

For minimizing $\mathcal{L}(a, b)$, we need

$$\nabla \mathcal{L} = \begin{pmatrix} \partial \mathcal{L} / \partial a \\ \partial \mathcal{L} / \partial b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Hence,

$$\frac{\partial \mathcal{L}}{\partial a} = -2 \sum_i \{y_i - ax_i - b\} x_i = 0 \quad (3)$$

$$\implies a \sum_i x_i^2 + b \sum_i x_i = \sum_i x_i y_i \quad (4)$$

Additionally,

$$\frac{\partial \mathcal{L}}{\partial b} = -2 \sum_i \{y_i - ax_i - b\} = 0 \quad (5)$$

$$\implies a \sum_i x_i + b \sum_i 1 = \sum_i y_i \quad (6)$$

Thus, in this case, estimating the coefficient pair a, b is a simple case for solving for the matrix

$$\begin{pmatrix} \sum_i x_i^2 & \sum_i x_i \\ \sum_i x_i & \sum_i 1 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum_i x_i y_i \\ \sum_i y_i \end{pmatrix}$$

Avoiding multicollinearity simply becomes a restriction of imposed non singularity on the matrix

$$M = \begin{pmatrix} \sum_i x_i^2 & \sum_i x_i \\ \sum_i x_i & \sum_i 1 \end{pmatrix}$$

In a strict theoretial sense that becomes

$$\sum_i x_i^2 \neq \frac{1}{n} \left\{ \sum_i x_i \right\}^2$$

That is only possible if all x_i values are the same, hence the dataset would be reduced by a dimension of at least one with respect to the geometry that it is trying to fit itself to, basically the definition of a singular matrix.

2.1.2 The multivariate function

Now, if the issue is extended to the multivariate case, herein one would have $y = a_1x_1 + a_2x_2 + \cdots + a_mx_m + a_{m+1} + \epsilon$ with $\epsilon \sim N(0, \sigma^2)$. Obviously the linear fit in this case is $f(x) = a_1x_1 + a_2x_2 + \cdots + a_mx_m + a_{m+1}$.

Thus the dataset in this case would be $(x_{1,i}, x_{2,i}, \cdots, x_{m,i}, y_i)$ with $i \in [1, n]$

In this case the objective would be $\mathcal{L}(a_1, a_2, \cdots, a_{m+1})$

$$\mathcal{L} = \sum_i \{a_1x_{1,i} + a_2x_{2,i} + \cdots + a_mx_{m,i} + a_{m+1} - y_i\}^2 \quad (7)$$

$$= \sum_{i=1}^n \left\{ a_{m+1} - y_i + \sum_{k=1}^m a_k x_{k,i} \right\}^2 \quad (8)$$

Thus for the gradient to be zero

$$\nabla \mathcal{L} = \begin{pmatrix} \partial \mathcal{L} / \partial a_1 \\ \partial \mathcal{L} / \partial a_2 \\ \vdots \\ \partial \mathcal{L} / \partial a_m \\ \partial \mathcal{L} / \partial a_{m+1} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}$$

That implies that $\forall p \in [1, m]$

$$\partial \mathcal{L} / \partial a_p = 2 \sum_{i=1}^n \{a_{m+1} - y_i + \sum_{k=1}^m a_k x_{k,i}\} x_{p,i} = 0 \quad (9)$$

$$\implies a_{m+1} \sum_{i=1}^n x_{p,i} + \sum_{i=1}^n x_{p,i} \left\{ \sum_{k=1}^m a_k x_{k,i} \right\} = \sum_{i=1}^n x_{p,i} y_i \quad (10)$$

$$\implies \sum_{k=1}^m a_k \sum_{i=1}^n x_{k,i} x_{p,i} + a_{m+1} \sum_{i=1}^n x_{p,i} = \sum_{i=1}^n x_{p,i} y_i \quad (11)$$

This means that the first m equations can be written down as shown below

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{1,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{1,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{1,i} \right) + a_{m+1} \sum_{i=1}^n x_{1,i} = \sum_{i=1}^n x_{1,i} y_i \quad (12)$$

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{2,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{2,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{2,i} \right) + a_{m+1} \sum_{i=1}^n x_{2,i} = \sum_{i=1}^n x_{2,i} y_i \quad (13)$$

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{3,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{3,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{3,i} \right) + a_{m+1} \sum_{i=1}^n x_{3,i} = \sum_{i=1}^n x_{3,i} y_i \quad (14)$$

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{4,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{4,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{4,i} \right) + a_{m+1} \sum_{i=1}^n x_{4,i} = \sum_{i=1}^n x_{4,i} y_i \quad (15)$$

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{5,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{5,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{5,i} \right) + a_{m+1} \sum_{i=1}^n x_{5,i} = \sum_{i=1}^n x_{5,i} y_i \quad (16)$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad = \quad \vdots \quad (17)$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad = \quad \vdots \quad (18)$$

$$a_1 \left(\sum_{i=1}^n x_{1,i} x_{m,i} \right) + a_2 \left(\sum_{i=1}^n x_{2,i} x_{m,i} \right) + \cdots + a_m \left(\sum_{i=1}^n x_{m,i} x_{m,i} \right) + a_{m+1} \sum_{i=1}^n x_{m,i} = \sum_{i=1}^n x_{m,i} y_i \quad (19)$$

$$(20)$$

However since in parameter space there are $m+1$ parameters $a_1, a_2, a_3, \dots, a_m, a_{m+1}$ hence $m+1$ conditions are needed wherein the $(m+1)$ th condition comes from setting the last minima $\partial \mathcal{L} / \partial a_{m+1} = 0$ as shown below

$$\partial \mathcal{L} / \partial a_{m+1} = 2 \sum_{i=1}^n \left\{ a_{m+1} - y_i + \sum_{k=1}^m a_k x_{k,i} \right\} = 0 \quad (21)$$

$$\implies \sum_{k=1}^m a_k \sum_{i=1}^n x_{k,i} + a_{m+1} \sum_{i=1}^n 1 = \sum_{i=1}^n y_i \quad (22)$$

In an expanded form this expands to the form shown below

$$a_1 \sum_{i=1}^n x_{1,i} + a_2 \sum_{i=1}^n x_{2,i} + \cdots + a_m \sum_{i=1}^n x_{m,i} + a_{m+1} \sum_{i=1}^n 1 = \sum_{i=1}^n y_i$$

With $m+1$ linear conditions in the parameters from a_1 to a_{m+1} , the $(m+1) \times (m+1)$ matrix representation of $[M]$ can be shown below

$$[M] = \begin{pmatrix} \sum_{i=1}^n x_{1,i}^2 & \sum_{i=1}^n x_{2,i} x_{1,i} & \cdots & \sum_{i=1}^n x_{m,i} x_{1,i} & \sum_{i=1}^n x_{1,i} \\ \sum_{i=1}^n x_{1,i} x_{2,i} & \sum_{i=1}^n x_{2,i}^2 & \cdots & \sum_{i=1}^n x_{m,i} x_{2,i} & \sum_{i=1}^n x_{2,i} \\ \sum_{i=1}^n x_{1,i} x_{3,i} & \sum_{i=1}^n x_{2,i} x_{3,i} & \cdots & \sum_{i=1}^n x_{m,i} x_{3,i} & \sum_{i=1}^n x_{3,i} \\ \sum_{i=1}^n x_{1,i} x_{4,i} & \sum_{i=1}^n x_{2,i} x_{4,i} & \cdots & \sum_{i=1}^n x_{m,i} x_{4,i} & \sum_{i=1}^n x_{4,i} \\ \sum_{i=1}^n x_{1,i} x_{5,i} & \sum_{i=1}^n x_{2,i} x_{5,i} & \cdots & \sum_{i=1}^n x_{m,i} x_{5,i} & \sum_{i=1}^n x_{5,i} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \sum_{i=1}^n x_{1,i} x_{m,i} & \sum_{i=1}^n x_{2,i} x_{m,i} & \cdots & \sum_{i=1}^n x_{m,i}^2 & \sum_{i=1}^n x_{m,i} \\ \sum_{i=1}^n x_{1,i} & \sum_{i=1}^n x_{2,i} & \cdots & \sum_{i=1}^n x_{m,i} & \sum_{i=1}^n 1 \end{pmatrix}$$

Similarly, the $(m+1) \times 1$ coefficient matrix $[C]$ can be drawn as shown below.

$$[C] = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_m \\ a_{m+1} \end{pmatrix}$$

To be able to complete the definition, the result vector also has to be defined as a $(m + 1) \times 1$ matrix $[B]$ as shown below

$$[B] = \begin{pmatrix} \sum_{i=1}^n x_{1,i} y_i \\ \sum_{i=1}^n x_{2,i} y_i \\ \sum_{i=1}^n x_{3,i} y_i \\ \vdots \\ \sum_{i=1}^n x_{m,i} y_i \\ \sum_{i=1}^n y_i \end{pmatrix}$$

With these definitions structured, now the relationship can be established

$$[M] \cdot [C] = [B]$$

Obviously the non-singularity of $[M]$ is necessary in order not to reduce its rank else the situation of multicollinearity can arise.